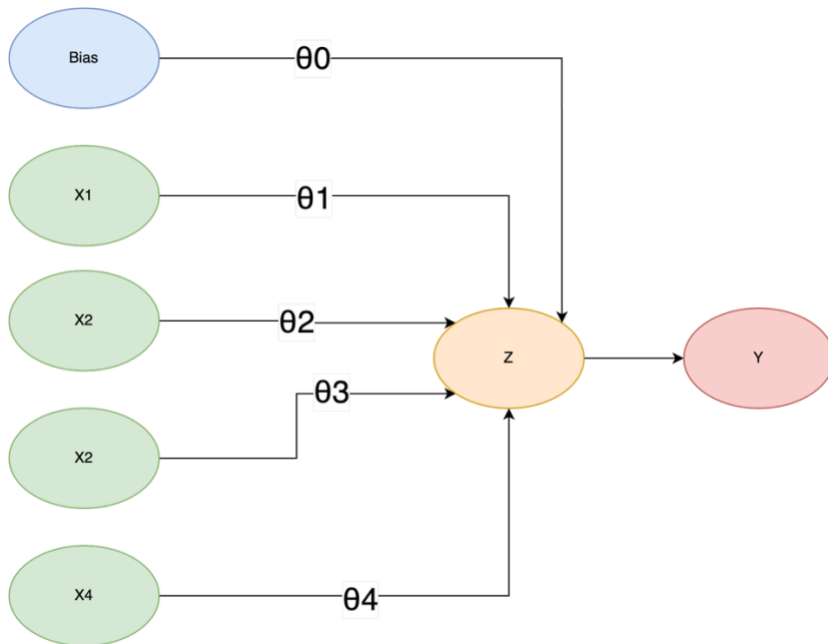
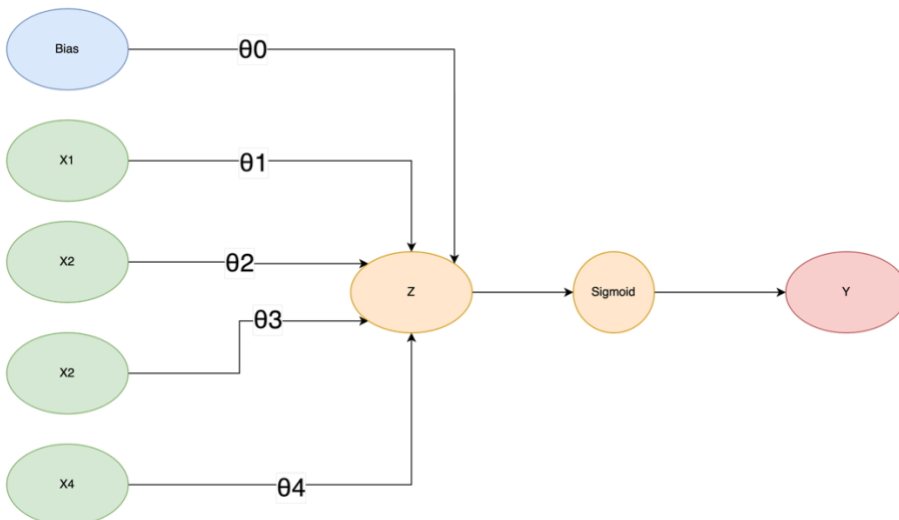


Evan Edelstein
EN.605.645.82.SP26
Module 13 – self check

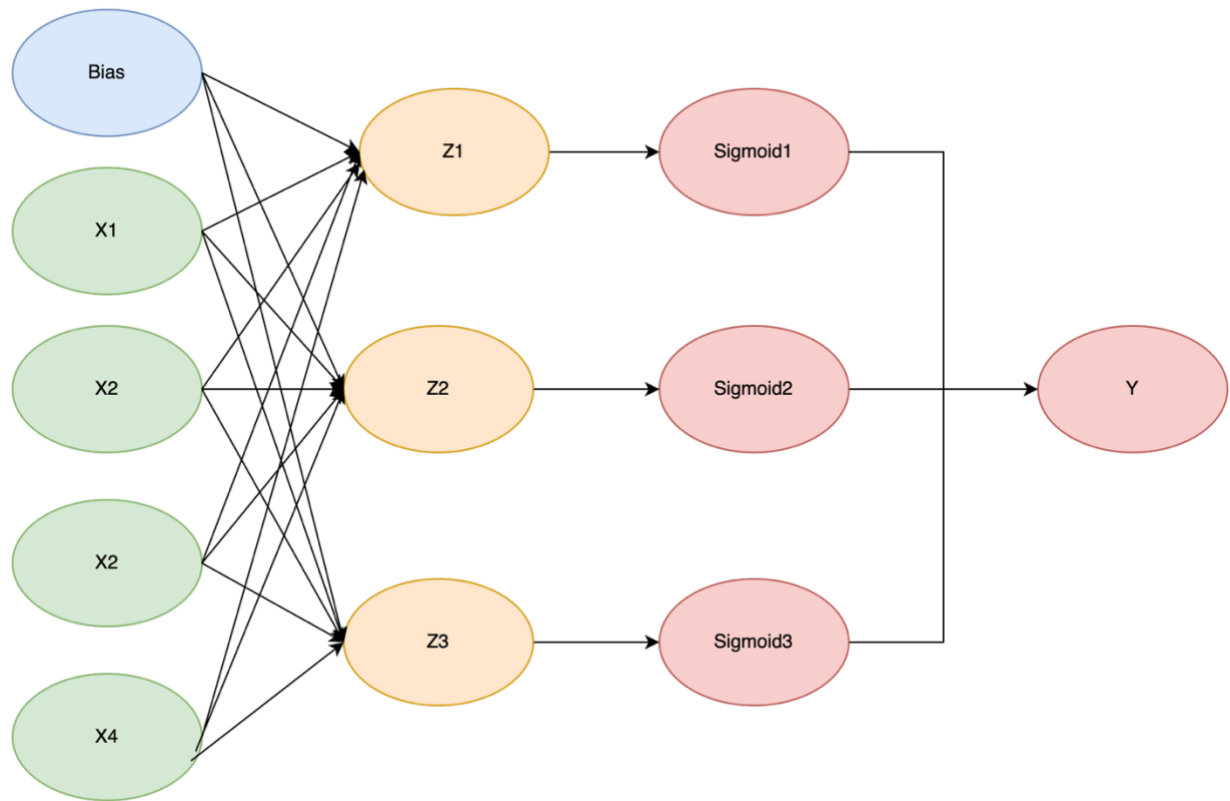
1)



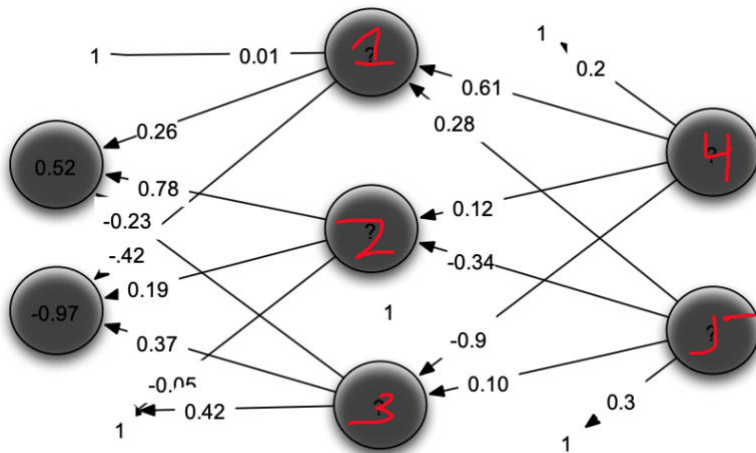
2) The difference between linear and logistic regression is the added step of applying the sigmoid function $f(z) = \frac{1}{1+e^{-z}}$ to z , the weighted sum of theta and x



3)



4)



Feed Forward:

Input: [0.52, -0.97]

$Z = \text{bias} + w_1 * x_1 + w_2 * x_2$

Sigmoid = $1/(1+e^{-z})$

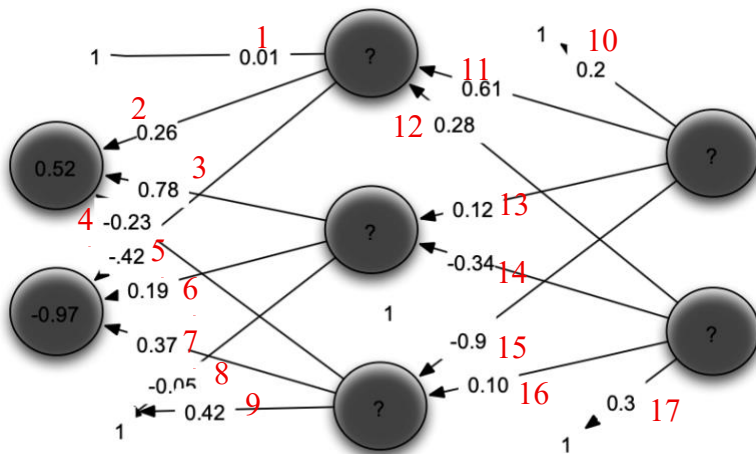
Delta:

Output: $\delta = \hat{y} (1 - \hat{y}) (y - \hat{y})$

$$\text{Hidden: } \delta = \hat{y} (1 - \hat{y}) \sum \theta_{i,o} \delta_{i,o}$$

Node	Bias	Weight1	Weight2	Weight3	Z	Sigmoid ($\hat{y}$)	Delta
1	0.01	0.26	-0.42		$0.01+0.26*0.52 + -0.42*-0.97 = 0.553$	$1/(1+e^{-0.553}) = 0.634$	$0.634 * (1-0.634) * (0.61*0.108 + 0.28 * -0.142) = 0.006$
2	-0.05	0.78	0.19		$-0.05+0.78*0.52+0.19*-0.97 = 0.1713$	$1/(1+e^{-0.171}) = 0.542$	$0.542 * (1-0.545) * (0.12*0.108 + -0.34*-0.142) = 0.015$
3	0.42	-0.23	0.37		$0.42+-0.23*0.52+0.37*-0.97 = -0.059$	$1/(1+e^{0.059}) = 0.485$	$0.485 * (1 - 0.485) * (-0.9*0.108 + 0.1*-0.142) = -0.028$
4	0.2	0.61	0.12	-0.9	$0.2 + 0.61*0.634 + 0.12*0.542 - 0.9*0.485 = 0.215$	$1/(1+e^{-0.215}) = 0.554$	$0.544 * (1 - 0.554) * (1 - 0.554) = 0.108$
5	0.3	0.28	-0.34	0.10	$0.3+ 0.28*0.634 - 0.34*0.542+0.1*0.485 = 0.342$	$1/(1+e^{-0.342}) = 0.585$	$0.585 * (1 - 0.585) * (0 - 0.585) = -0.142$

$$w_{+} = a * \delta * x$$



Weight	Old	Update
1	0.01	$0.01+0.01*0.006 *1 = 0.01$
2	0.26	$0.26+0.01*0.006*0.52=0.260$
3	0.78	$0.78+0.01*0.015*0.52 = 0.780$
4	-0.23	$-0.23+0.01*-0.028*-0.52 = -0.230$
5	-0.42	$-0.42+0.01*0.006*-0.97 = -0.420$
6	0.19	$0.19+0.01*0.015*-0.97 = 0.190$
7	0.37	$0.37+0.01*-0.028*-0.97 = 0.37$

8	-0.05	$-0.05+0.01*0.015*1 = -0.0498$
9	0.42	$0.42+0.01*-0.028*1 = 0.4197$
10	0.2	$0.2+0.01*0.006+1=0.2006$
11	0.61	$0.61+0.01*0.108*0.634 = 0.6106$
12	0.28	$0.28+0.01*0.634*-0.142 = 0.279$
13	0.12	$0.12+0.01*0.108 * 0.542=0.1206$
14	-0.34	$-0.34+0.01*0.542*-0.142=-0.34077$
15	-0.9	$-0.9+0.01*0.485*0.108 = -0.899$
16	0.10	$0.1+0.01*0.485*-0.142=0.0993$
17	0.3	$0.3+0.01*-0.142*1= 0.298$